Data Mining Project Report

Title

CFPB Consumer Complaints: Predicting Monetary Relief Outcomes from Complaint-Time Information

Names

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Course Title

BA 360: Business Data Mining

Contents

1 Dataset Description

1.1 General Context

Describe the business and regulatory context of CFPB complaints.

1.2 Data Source

- Source: CFPB Consumer Complaint Database
- Link: https://www.consumerfinance.gov/data-research/consumer-complaints/
- File used in project: `data/processed/complaints_sample.csv`

1.3 Unit of Observation

Explain what one row represents.

1.4 Main Variables

Summarize key variables used in EDA and modeling.

2 Problem Statement

2.1 Problem to Study

State the analytical problem clearly.

2.2 Objective

Define the objective of the analysis.

2.3 Research Questions

List the main research questions.

2.4 Target Variable

Explain the binary target design (monetary relief vs no monetary relief) and leakage-safe constraints.

3 Data Preparation

3.1 Missing Values

Explain handling strategy and justification.

3.2 Outliers and Quality Checks

Describe checks and decisions.

3.3 Feature Transformations

Detail text cleaning, vectorization, and any transformations.

3.4 Encoding and Scaling

Document categorical encoding and scaling choices.

3.5 Feature Selection and Leakage Control

List excluded post-resolution fields and explain why.

4 Methods and Analysis

4.1 Modeling Strategy

Explain train/test setup and model comparison approach.

4.2 Models Implemented

- Logistic Regression (TF-IDF)
- Naive Bayes (TF-IDF)
- KNN (SVD-reduced text)
- Random Forest (structured + text)
- MLP Neural Network
- Voting Classifier Ensemble

4.3 Why These Methods

Justify method choices for this problem.

5 Results and Interpretation

5.1 Model Comparison

Insert summary table from reports outputs.

5.2 Threshold Analysis

Discuss trade-offs and operational threshold recommendation.

5.3 Confusion Matrix Insights

Interpret false positives and false negatives in business terms.

5.4 Key Findings

Provide clear, practical conclusions.

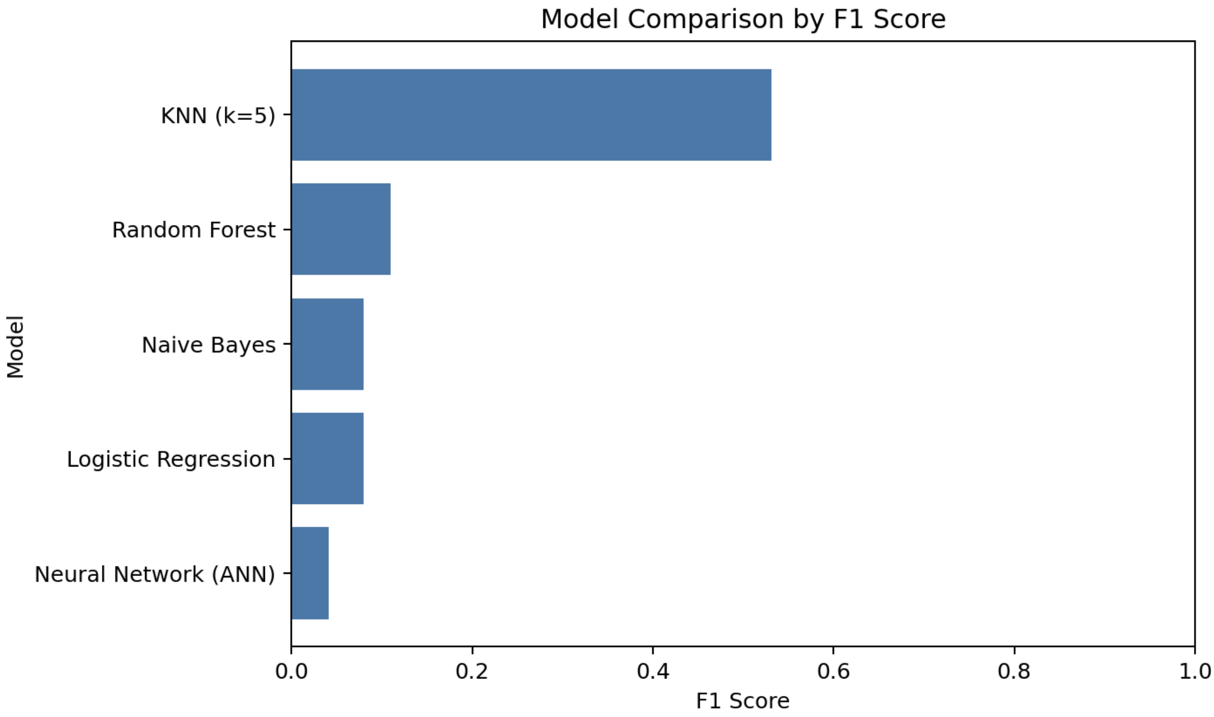

Figure 1: Model comparison table figure

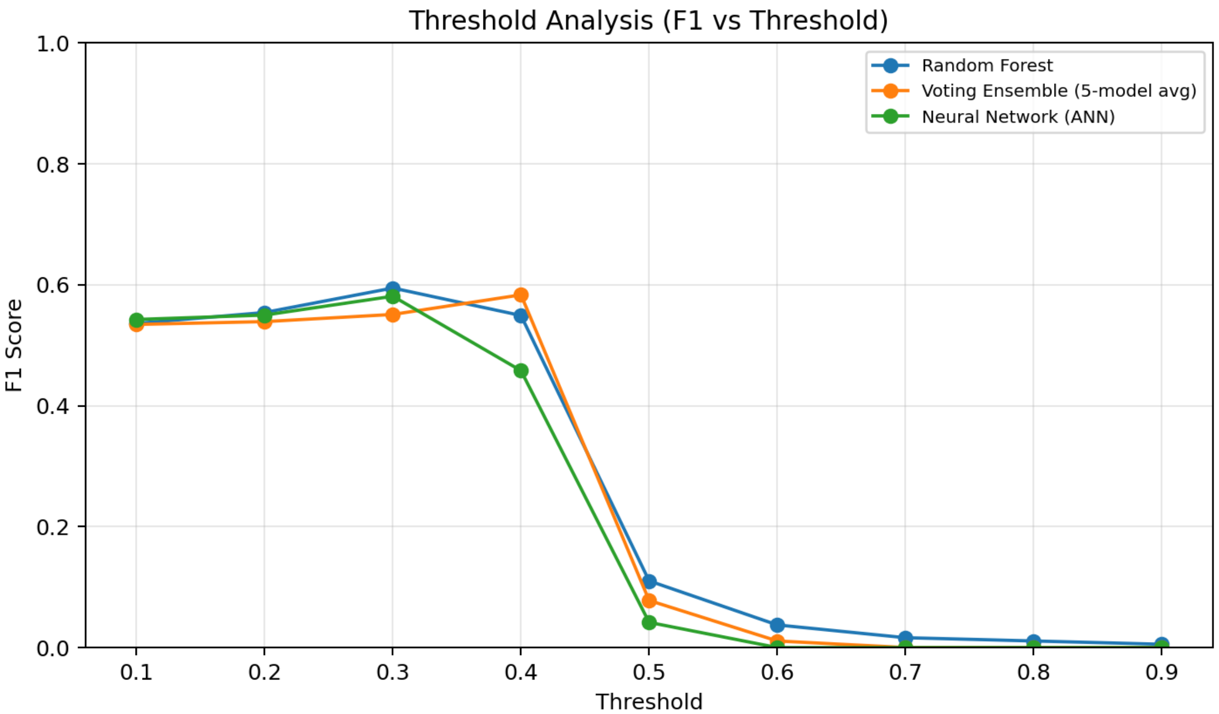

Figure 2: Threshold analysis

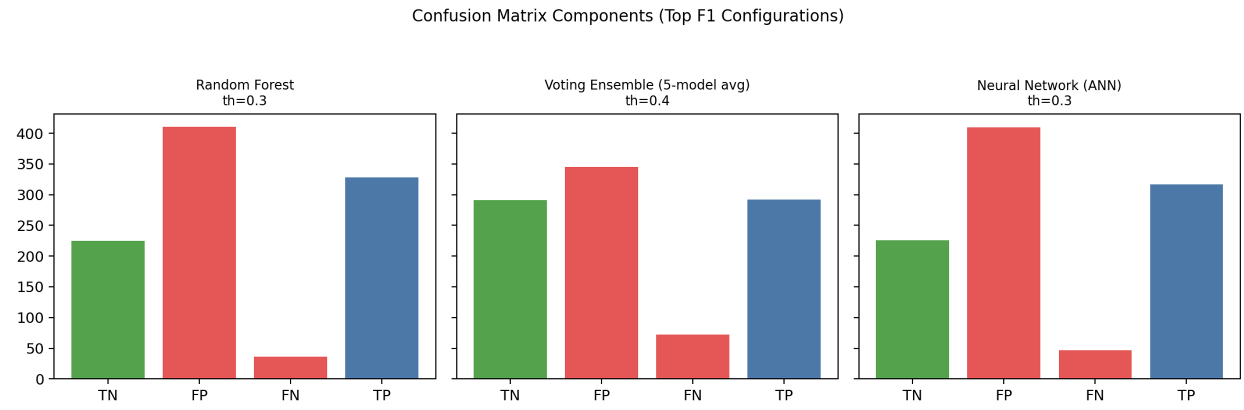

Figure 3: Confusion matrix

6 Conclusion

6.1 Objective Recap

Summarize initial objective.

6.2 Methods Recap

Summarize methods used.

6.3 Main Results

Summarize strongest findings and selected model.

6.4 Business Insights

Summarize actionable recommendations.

7 Appendix (Optional)

7.1 Evolution and Corrections

Reference the mistakes-to-improvements timeline.

7.2 Additional Tables

Add detailed model metrics if needed.